Experiment 8: Rayleigh–Taylor Instability (SIMPLE + VOF)

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1 Aim

Simulate the Rayleigh–Taylor instability of a heavy fluid resting above a light fluid using a pressure-based SIMPLE solver coupled with a Volume-of-Fluid (VOF) transport equation, and observe the interface evolution from $t = 0$ to $t = 6$ s.

2 Assumptions & Limitations

- **2D, Laminar, Unsteady:** Flow is two-dimensional, laminar and time-accurate; no turbulence model is used.
- **Two Incompressible, Immiscible Fluids:** Each fluid is incompressible ($\nabla \cdot \mathbf{u} = 0$) with constant density; density varies only through the volume fraction α.
- **Constant Viscosity:** A single dynamic viscosity μ is used for the mixture; no surface tension.
- **Boussinesq-type Buoyancy:** Gravity acts in $-y$ and only the density difference from a reference density, $(\rho - \rho_{ref})g$, is retained as body force (the hydrostatic part is absorbed in the pressure).
- **First-order VOF:** The interface is advected by a bounded first-order upwind scheme, hence it is numerically diffused (no geometric reconstruction/PLIC).
- **Collocated Grid:** u, v, p, α share one $(n_y \times n_x)$ mesh; no Rhie–Chow interpolation is used.

3 Governing Equations & Simplification

Volume fraction and mixture density. $\alpha = 1$ denotes the heavy fluid (top) and $\alpha = 0$ the light fluid:

$$\rho = \rho_l + (\rho_h - \rho_l)\,\alpha, \qquad At = \frac{\rho_h - \rho_l}{\rho_h + \rho_l} \tag{1}$$

Continuity and momentum (variable density, constant μ):

$$\nabla \cdot \mathbf{u} = 0, \qquad \frac{\partial(\rho \mathbf{u})}{\partial t} + \nabla \cdot (\rho \mathbf{u}\mathbf{u}) = -\nabla p + \mu \nabla^2 \mathbf{u} - (\rho - \rho_{ref})\, g\, \hat{\mathbf{j}} \tag{2}$$

with $\rho_{ref} = \frac{1}{2}(\rho_l + \rho_h)$. Subtracting $\rho_{ref} g$ simply redefines p (removes the hydrostatic mean), so the only body force left is the *net* buoyancy that drives the instability. In components:

$$\frac{\partial(\rho u)}{\partial t} + \frac{\partial(\rho u u)}{\partial x} + \frac{\partial(\rho v u)}{\partial y} = -\frac{\partial p}{\partial x} + \mu \nabla^2 u \tag{3}$$

$$\frac{\partial(\rho v)}{\partial t} + \frac{\partial(\rho u v)}{\partial x} + \frac{\partial(\rho v v)}{\partial y} = -\frac{\partial p}{\partial y} + \mu \nabla^2 v - (\rho - \rho_{ref}) g \tag{4}$$

VOF transport. Because $\nabla \cdot \mathbf{u} = 0$, the conservative and advective forms coincide:

$$\frac{\partial \alpha}{\partial t} + \nabla \cdot (\mathbf{u}\alpha) = \frac{\partial \alpha}{\partial t} + u \frac{\partial \alpha}{\partial x} + v \frac{\partial \alpha}{\partial y} = 0 \tag{5}$$

Reference (linear theory). For a small single-mode perturbation of wavelength $\lambda = L_x$ the inviscid growth rate is $\sigma = \sqrt{At\, g\, k}$ with $k = 2\pi/\lambda$; the small Atwood number ($At \approx 0.091$) makes the growth slow and the roll-up gradual.

4 Discretization & Algorithm

4.1 Finite-Volume Momentum Equation

Integrating over a control volume $\Delta x \Delta y$ with implicit (backward) Euler in time gives, for each component,

$$a_P\, \phi_P^* = a_E \phi_E^* + a_W \phi_W^* + a_N \phi_N^* + a_S \phi_S^* + b_\phi + (\text{pressure term}), \qquad a_P = \sum a_{nb} + a_P^0 \tag{6}$$

where $a_P^0 = \rho_P \Delta x \Delta y / \Delta t$. Face fluxes and diffusion conductances are

$$F_f = (\rho u)_f A_f, \qquad D_f = \frac{\mu A_f}{\Delta}, \qquad \mathrm{Pe}_f = \frac{F_f}{D_f} \tag{7}$$

and the neighbour coefficients use the **Power-Law scheme** (Patankar):

$$a_E = D_e\, A(|\mathrm{Pe}_e|) + \max(-F_e, 0), \qquad a_W = D_w\, A(|\mathrm{Pe}_w|) + \max(F_w, 0) \tag{8}$$

$$A(\mathrm{Pe}) = \max\big[0,\ (1 - 0.1|\mathrm{Pe}|)^5\big] \tag{9}$$

(and similarly for a_N, a_S). Face values of u, v, ρ are linear averages of the two adjacent nodes (uniform grid). The source terms are

$$b_u = a_P^0 u^n, \qquad b_v = a_P^0 v^n - (\rho_P - \rho_{ref})\, g\, \Delta x \Delta y \tag{10}$$

and the pressure gradient is a central difference over two cells:

$$u_P^* \ni \frac{p_W - p_E}{2} \Delta y / a_P, \qquad v_P^* \ni \frac{p_S - p_N}{2} \Delta x / a_P \tag{11}$$

4.2 Pressure Correction (SIMPLE)

Writing $u = u^* + u'$, $p = p^* + p'$ and neglecting neighbour-velocity corrections (the SIMPLE approximation):

$$u'_P = d_e \frac{p'_W - p'_E}{2}, \quad v'_P = d_n \frac{p'_S - p'_N}{2}, \qquad d_e = \frac{\Delta y}{a_P},\ d_n = \frac{\Delta x}{a_P} \tag{12}$$

Imposing continuity gives the pressure-correction equation

$$a'_P p'_P = a'_E p'_E + a'_W p'_W + a'_N p'_N + a'_S p'_S + b, \qquad a'_E = \rho_P d_e \Delta y,\ a'_N = \rho_P d_n \Delta x \tag{13}$$

$$b = \rho_P \frac{u^*_W - u^*_E}{2}\Delta y + \rho_P \frac{v^*_S - v^*_N}{2}\Delta x \quad \text{(mass imbalance)} \tag{14}$$

p' is solved with 50 Jacobi sweeps and zero-gradient walls. The residual reported is $\max|b|$.

4.3 Under-relaxation and Correction

Pressure is under-relaxed, $p = p^* + \alpha_p p'$ with $\alpha_p = 0.3$, while velocities are corrected with the full p'. The implicit transient term a^0_P itself acts as inertial relaxation on the velocities.

4.4 VOF Update

The advection of α is explicit and first-order upwind (bounded):

$$\alpha^{n+1}_P = \alpha^n_P - \Delta t \left(u_P \left.\frac{\partial \alpha}{\partial x}\right|_{up} + v_P \left.\frac{\partial \alpha}{\partial y}\right|_{up} \right), \qquad \alpha \in [0,1] \text{ (clipped)} \tag{15}$$

followed by the density update $\rho = \rho_l + (\rho_h - \rho_l)\alpha$.

4.5 Adaptive Time Step

Since $\mathbf{u} = 0$ at $t = 0$, a convective limit alone gives $\Delta t \to \infty$, so a three-way minimum is used:

$$\Delta t = \min\left[\frac{CFL_{max}}{|u|_{max}/\Delta x + |v|_{max}/\Delta y},\ \frac{0.45}{2\nu_{max}(1/\Delta x^2 + 1/\Delta y^2)},\ 0.01 \right], \quad \nu_{max} = \frac{\mu}{\rho_l} \tag{16}$$

with $CFL_{max} = 0.3$. The viscous bound uses the *light* fluid because it has the largest kinematic viscosity.

4.6 Solution Procedure per Time Step

1. Compute Δt; store $\mathbf{u}^n$, α^n.
2. Repeat $n_{inner} = 3$ times: solve $u^*, v^* \to$ apply wall BCs $\to$ solve $p' \to$ update $p \to$ correct $u, v \to$ apply BCs.
3. Advect α with the corrected velocity and update ρ.
4. Advance t; store snapshots at $t = 0, 1.5, 3, 4.5, 6$ s.

5 Methodology and Boundary Conditions

Parameter	Value	Parameter	Value
Domain ($L_x \times L_y$)	1×2	Grid ($n_x \times n_y$)	40×80
Light density ρ_l	1.0	Heavy density ρ_h	1.2
Atwood number At	0.091	Gravity g	1.0
Viscosity μ	2×10^{-3}	Inner SIMPLE iterations	3
Pressure relaxation α_p	0.3	End time t_{end}	6 s
CFL_{max}	0.3	Δt_{max}	0.01

Table 1: Simulation parameters (non-dimensional units).

Initial condition: fluid at rest, with a single-mode interface $y_{int} = \frac{L_y}{2} + 0.05 L_x \cos(2\pi x / L_x)$ and the heavy fluid above it ($\alpha = 1$ for $y > y_{int}$).

Boundary conditions:

- **All four walls:** no-slip, $u = v = 0$.
- **Pressure, pressure correction and α:** zero normal gradient on all walls.

6 Results & Discussions

Figure 1 shows the α field at $t = 0$, 1.5, 3, 4.5 and 6 s. The initial cosine perturbation grows under buoyancy: heavy fluid sinks as a spike/bubble pair, and the interface progressively distorts, as expected for the Rayleigh–Taylor instability at low Atwood number. The interface is slightly diffuse due to the first-order upwind VOF scheme.

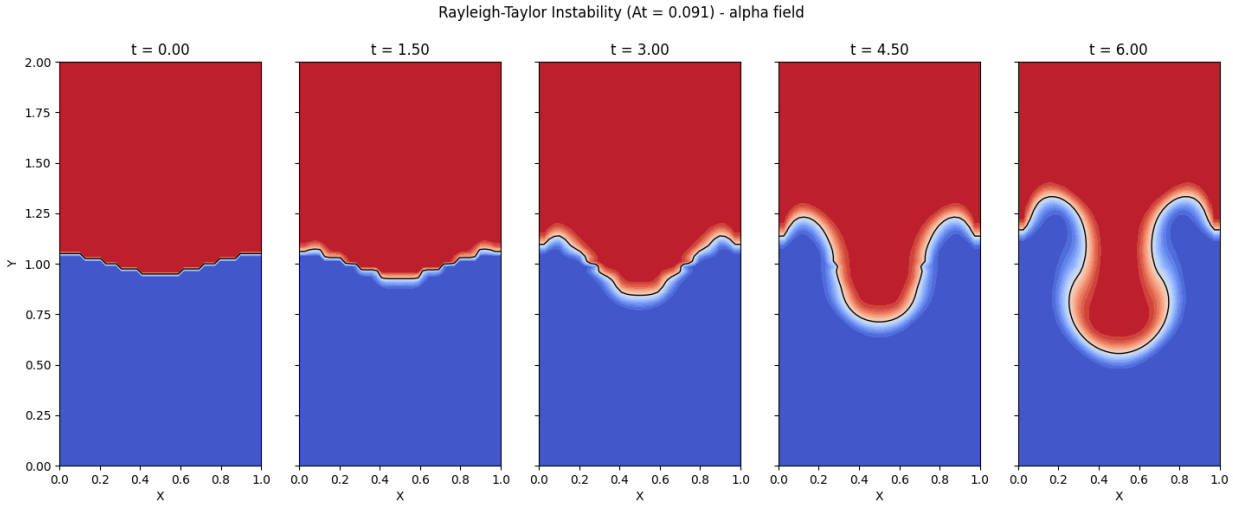


Figure 1: Volume-fraction (α) field at $t = 0, 1.5, 3.0, 4.5, 6.0$ s with the $\alpha = 0.5$ interface contour.

Figure 2 shows the streamlines at the final time. The flow organizes into counter-rotating recirculation cells on either side of the descending heavy-fluid spike and rising light-fluid bubble, with the highest velocity magnitudes near the interface where the density jump drives the motion.

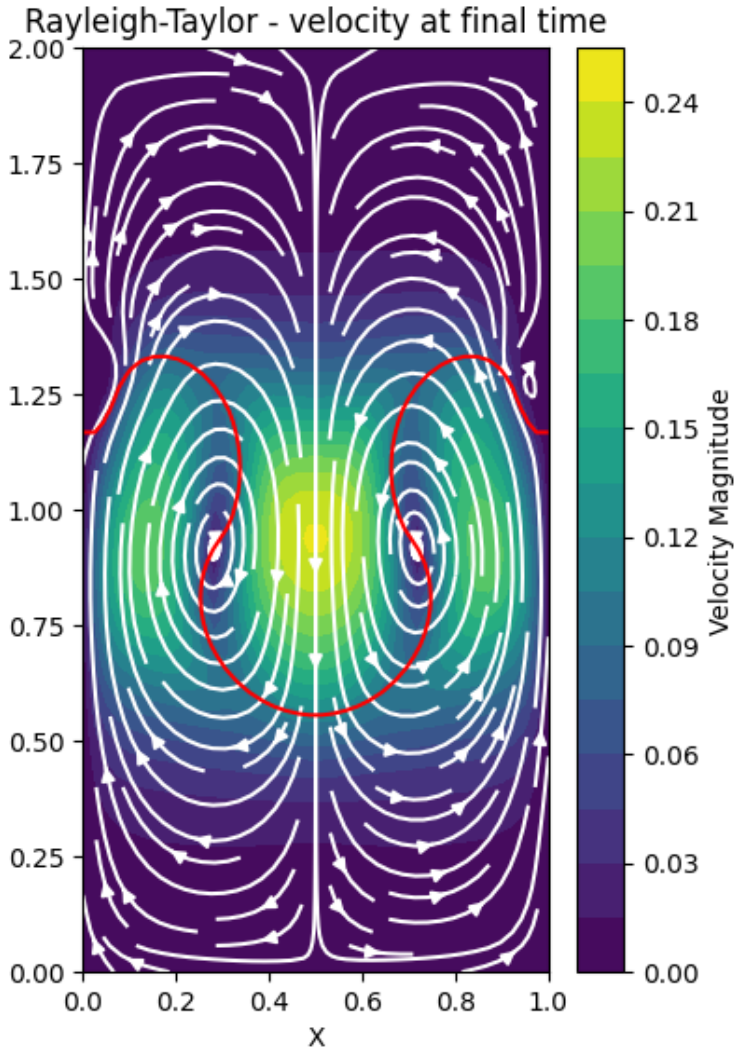


Figure 2: Velocity magnitude and streamlines at the final time with the interface in red.

7 Conclusion

A SIMPLE-based collocated solver with Power-Law convection–diffusion discretization was coupled with an explicit bounded VOF advection to capture the Rayleigh–Taylor instability. The adaptive time step (convective, viscous and capped limits) handled the zero-velocity

start, and the interface evolution is physically consistent. Future improvements include Rhie–Chow interpolation, higher-order or geometric (PLIC) interface advection, and a finer grid to sharpen the interface.